

Question Paper

Exam Date & Time: 26-Nov-2025 (02:00 PM - 05:00 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

MANIPAL INSTITUTE OF TECHNOLOGY - BENGALURU
FIFTH SEMESTER B.TECH. DEGREE EXAMINATIONS - NOVEMBER 2025

WEB AND SOCIAL MEDIA ANALYTICS [DS_3153]

Marks: 50

Duration: 180 mins.

PART-A

Answer all the questions.

- 1) A multinational retail company, GlobalMart, faces challenges in making timely decisions due to data (5)
being scattered across departments and regions. Store managers often rely on outdated sales
reports, while the marketing team struggles to coordinate promotional campaigns across countries.
A) To address this, GlobalMart implements an integrated Decision Support System (DSS) that
includes:

- A centralized data warehouse for all sales and inventory data.
- Online Analytical Processing (OLAP) tools to analyze customer trends.
- Web-based dashboards accessible from mobile devices.
- Collaboration platforms for managers and teams across different time zones.

Analyse how the integration of these technologies can enhance GlobalMart's decision-making process. In your answer, explain how each component supports different stages or aspects of decision making

Course Outcome: "CO2" / Cognitive Level: "Analysis"

- B) The hospital's management wants to improve its emergency response efficiency. They are (3)
considering adopting either a DSS or a BI tool. Apply your knowledge to identify which system
would provide more actionable support during critical decision-making situations and justify your
choice with reasoning

Course Outcome: "CO2" / Cognitive Level: "Application"

- C) A food delivery company wants to estimate how many orders it will get during the Diwali week to (2)
plan delivery staff schedules. Choose which type of analytics should it use and why?

Course Outcome: "CO2" / Cognitive Level: "Application"

- 2) A financial firm wants to automatically monitor customer chat transcripts for unprofessional or (5)
negative interactions by service associates.

- A) Using the sentence:
"The associate continuously provided unpleasant, reassuringly bad language."
Write Python code to apply tokenization, stop word removal, POS tagging, and lemmatization.
Interpret the output and explain how these preprocessing steps prepare the data for Text Mining
tasks such as identifying communication quality issues or behavioural patterns.

Course Outcome: "CO1" / Cognitive Level: "Application"

- B) Using the output from Q2a, demonstrate using Python code how these cleaned tokens can be (3)
transformed into a bag-of-words or TF-IDF matrix.

Explain how dimensionality reduction achieved through preprocessing helps the system detect patterns of unprofessional language more effectively.

Course Outcome: "CO1" / Cognitive Level: "Application"

- C) Explain Sentiment Analysis and list any two common tools or libraries used to perform it. Describe how sentiment scores can help flag negative or unprofessional customer interactions. (2)

Course Outcome: "CO1" / Cognitive Level: "Understand"

- 3) A smartphone manufacturing company wants to analyze customer feedback collected from various e-commerce platforms to improve its product quality. The dataset contains thousands of text reviews such as: (5)

A)

- "The display quality is amazing and very bright."
- "Battery drains too quickly."
- "Camera performance is outstanding."

The company's data analytics team plans to build a sentiment classification model that identifies each review as Positive (P) or Negative (N).

Demonstrate the steps involved in performing N-P polarity classification for this dataset.

Show how special lexicons are used in identification of sentiment polarity.

Differentiate between two common methods for polarity identification considering the above text

Course Outcome: "CO1" / Cognitive Level: "Application"

- B) Analyse the significance of multistep process of sentiment analysis with a diagram. (3)

Course Outcome: "CO1" / Cognitive Level: "Analysis"

- C) Explain the sources of data for sentiment analysis and list the common challenges that sentiment analysis has to deal with? (2)

Course Outcome: "CO1" / Cognitive Level: "Understand"

- 4) A business aiming for successful Search Engine Optimization (SEO) attempts to boost its organic ranking by solely maximizing keyword usage across all its pages, aiming for the highest possible Term-Frequency divided by Inverse-Document-Frequency (TF/IDF) score during the search engine's indexing phase. (5)

A)

Analyze the limitations of this content-focused strategy by differentiating the roles of Web Content Mining (WCM) and Web Structure Mining (WSM) in determining a page's final ranking. Explain how, despite maximizing its WCM/TF/IDF score, the page may still fail to achieve high visibility if it lacks the authority conferred by WSM-derived concepts, such as acting as an authoritative page (being pointed to) or leveraging highly specialized "Hubs"(pointing to authorities)

Course Outcome: "CO2" / Cognitive Level: "Analysis"

- B) Review of the GA4 Traffic Acquisition report shows that: (3)

- "Referral" traffic has high engaged sessions but low average engagement time.
- "Direct" traffic has fewer sessions but high conversion rates. Apply GA4 analysis skills to:

1. Interpret the likely user behaviour behind these patterns, and

2. Suggest one GA4 configuration or report setup that can confirm your interpretation.

Course Outcome: "CO3" / Cognitive Level: "Application"

- C) A company's analytical report shows high engagement but low conversion rates from Instagram. Classify "Best Practices in Social Media Analytics" to suggest one improvement strategy. (2)

- 5) A city government is planning to choose the most suitable location for constructing a new public hospital. The urban planning team has shortlisted four potential sites: (5)

A)

Site A: Close to the city center but expensive
 Site B: Near highways, moderate cost
 Site C: Low cost but far from main population zones
 Site D: Balanced location but limited land area

The team must evaluate these alternatives based on multiple criteria, including: Accessibility (distance from population centers), Land cost, Environmental impact, Future expansion potential, Availability of infrastructure.
 Demonstrate how Analytic Hierarchy Process (AHP), help in solving multi-criteria decision-making problems like site selection?

Describe the main steps involved in applying AHP to the above case - from defining the goal to determining the best alternative.

- B) Differentiate between Sensitivity Analysis and What-If Analysis in the context of the TechFab Industries case. (3)

Explain how the finance team can use these analyses to make better production and pricing decisions and Discuss how the insights from these analyses help *TechFab Industries* reduce risk and improve strategic planning before launching the new product.

- C) A financial analyst at an automobile company studies the relationship between production cost and profit. In one situation, the analyst changes the production cost to see how profit varies; in another, they set a desired profit and let the model find the necessary production cost. (2)
 Using this case, explain the difference between What-If Analysis and Goal Seeking, focusing on which variables are changed and which results are observed in each approach

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Event Name	Subject
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058-END SEMSTER-ODD-2025 DS_3153:WEB AND SOCIAL MEDIA ANALYTICS

#	Course Outcome	Description	Marks
1	1	CO 1	20
2	2	CO 2	15
3	3	CO 3	5
4	4	CO 4	10
Total			50

* **NA** - CO Not Selected

Event Name	Subject
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#	Level	No of Questions	Marks (%)
1	Remember	0	0(0%)

2	Understand	3	6(12%)
3	Application	7	26(52%)
4	Analysis	5	18(36%)
5	Create	0	0(0%)
6	Evaluate	0	0(0%)
	Total	15	50 (100%)